

1. Calcule a corrente i no resistor R5 da Figura 1, usando superposição.

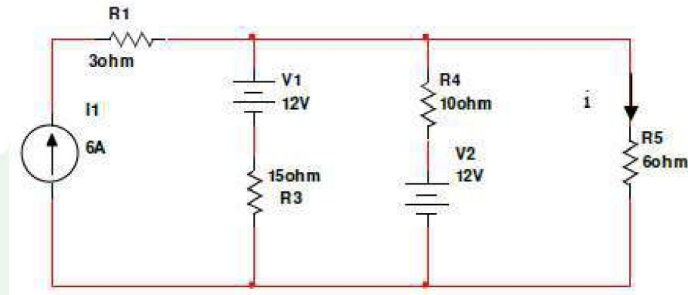


Figura 1

$$\frac{V_1 - 12}{15} + \frac{V_1}{10} + \frac{V_1}{6} = 0$$

$$2V_1 - 24 + 3V_1 + 5V_1 = 0$$

$$(2 + 3 + 5)V_1 = 24$$

$$10V_1 = 24$$

$$V_1 = 2,4V$$

$$i_1 = \frac{2,4}{6} = 0,4A$$

$$\frac{V_2}{15} + \frac{V_2 - 12}{10} + \frac{V_2}{6} = 0$$

$$2V_2 + 3V_2 - 36 + 5V_2 = 0$$

$$(2 + 3 + 5)V_2 = 36$$

$$10V_2 = 36$$

$$V_2 = 3,6V$$

$$i_2 = \frac{3,6}{6} = 0,6A$$

$$-6 + \frac{V_3}{15} + \frac{V_3}{10} + \frac{V_3}{6} = 0$$

$$-180 + 2V_3 + 3V_3 + 5V_3 = 0$$

$$(2 + 3 + 5)V_3 = 180$$

$$V_3 = \frac{180}{10} = 18V$$

$$i_3 = \frac{18}{6} = 3A$$

$$i = i_1 + i_2 + i_3 = 0,4 + 0,6 + 3 = 4A$$